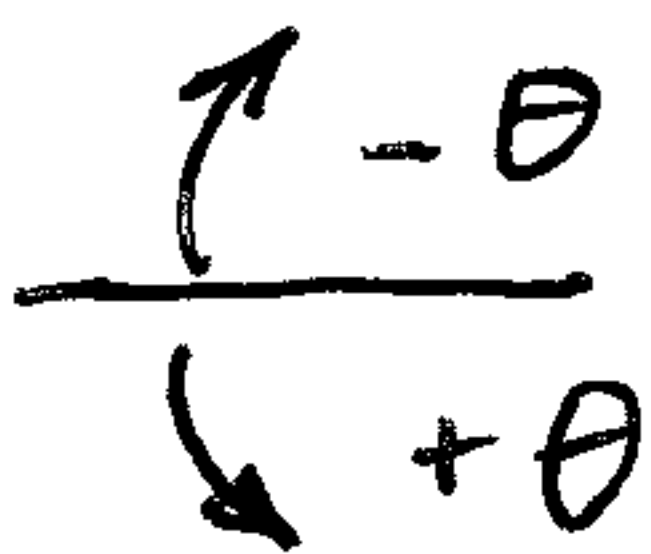
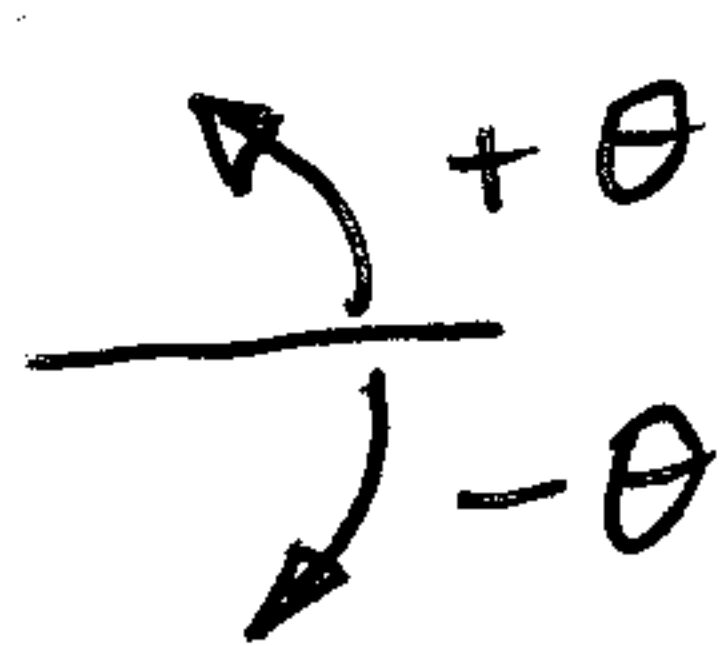


The θ_p Equation always gives the angle
From the I_x orientation To the nearest
 principal I (I_{max} or I_{min}). It also gives
 the correct sense or direction:



$$CW = -\theta$$

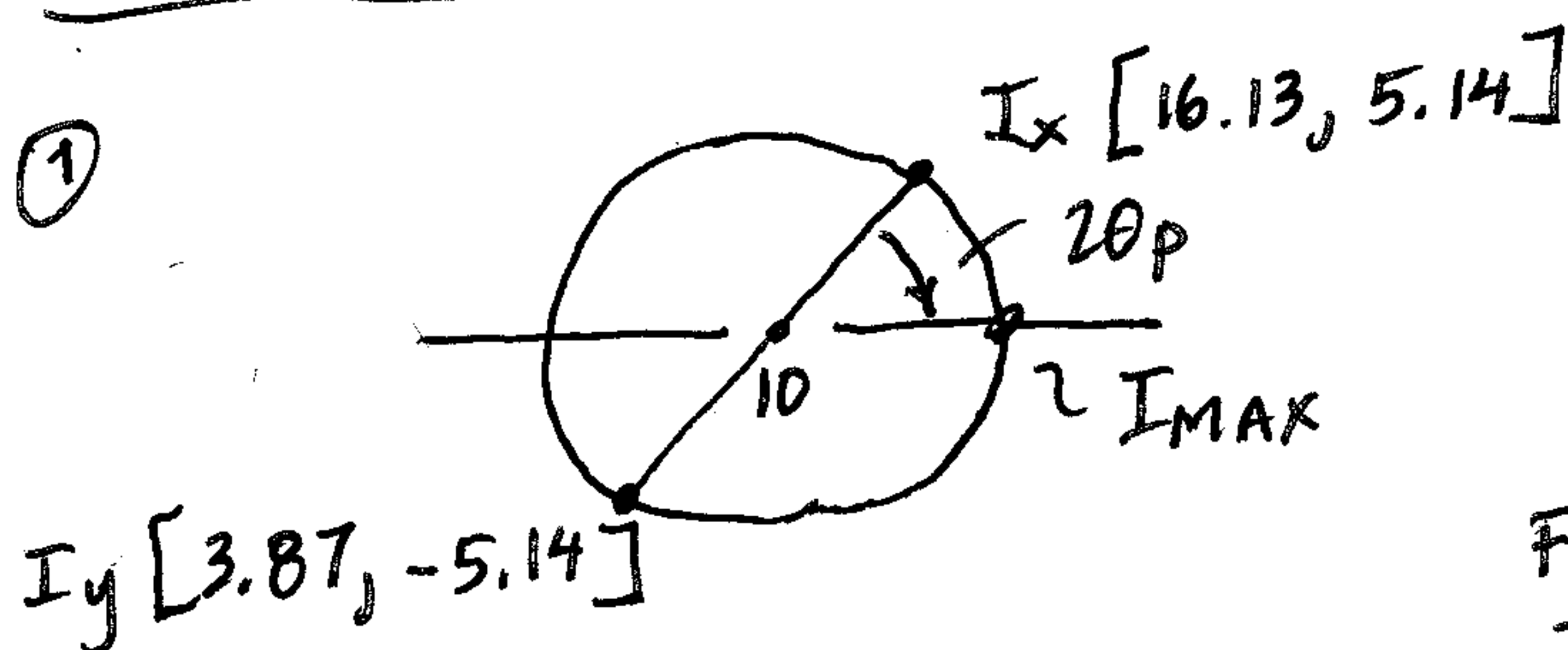
$$CCW = +\theta$$

θ_{p1} is angle
 from x
 to I_{max}
 θ_{p2} is angle
 to I_{min}

The θ_p equation:

$$\theta_p = \frac{1}{2} \tan^{-1} \frac{-I_{xy}}{\left(\frac{I_x - I_y}{2}\right)}$$

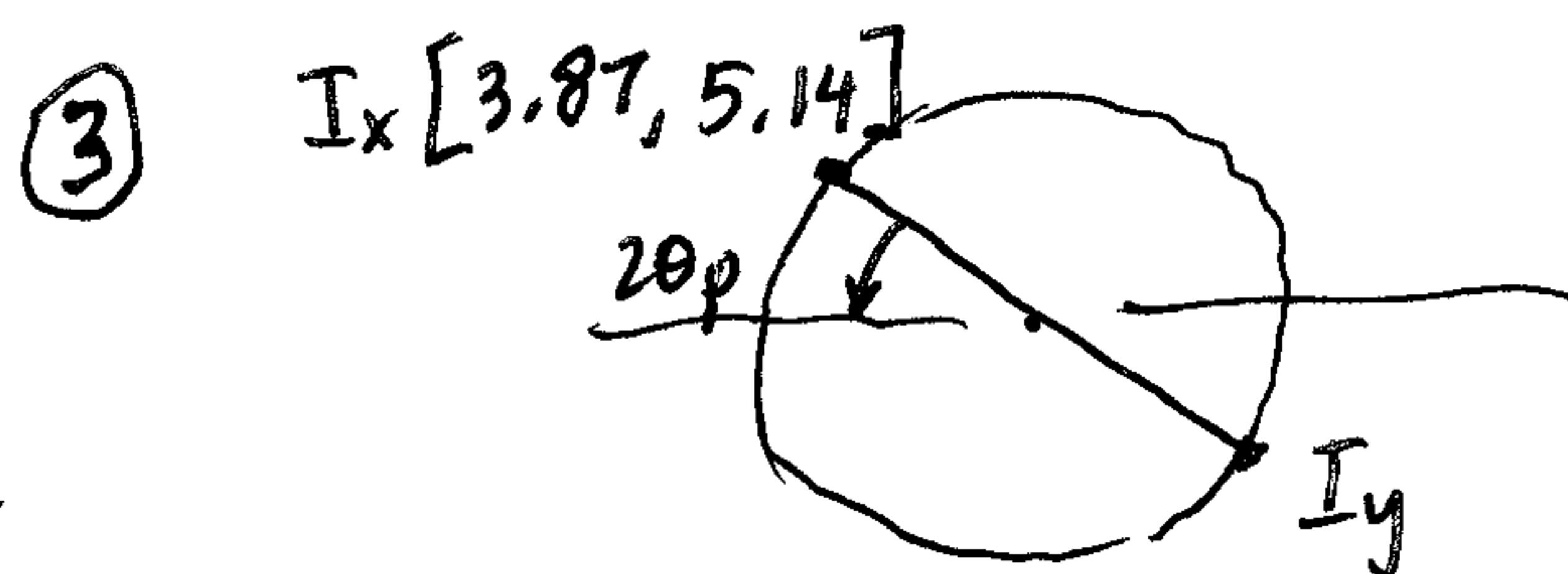
Four cases (made up)



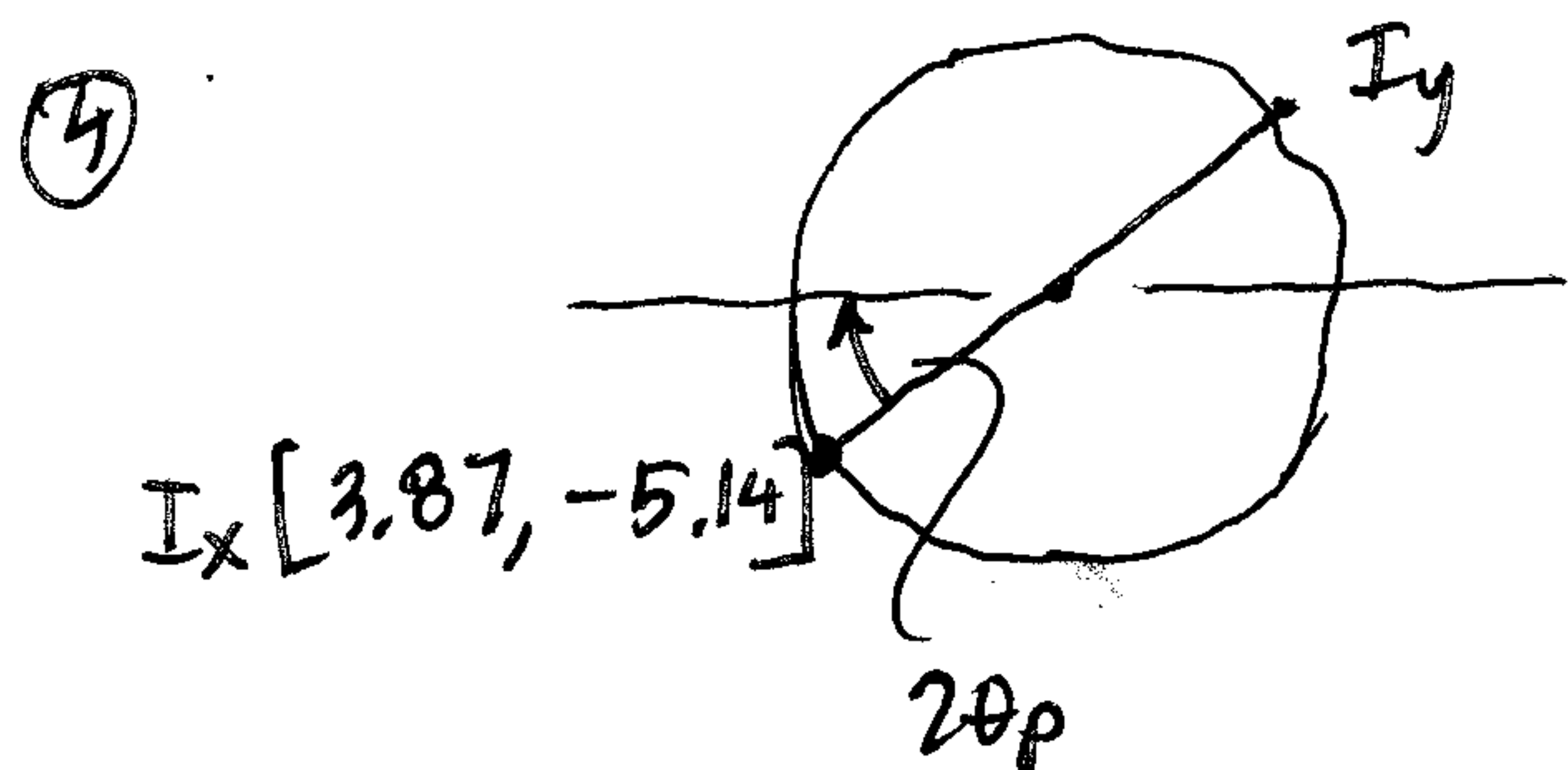
Calculate:
 $2\theta_p$ (on MC) = -40°
 θ_p (on Area) = -20°
From I_x to I_{max} (This is θ_{p1})



$2\theta_p = +40^\circ$
 $\theta_p = +20^\circ$
 To I_{max} . (θ_{p1})



$2\theta_p = +40^\circ$
 $\theta_p = +20^\circ$
 To I_{min} . (θ_{p2})



$2\theta_p = -40^\circ$
 $\theta_p = -20^\circ$
 To I_{min} . (θ_{p2})